Schur Covariance Evaluation

A Principled Pseudo-Likelihood in High Dimensions

Peter Cotton

Draft — `microprediction/precise`

Abstract

Ask a covariance matrix how surprised it is by fresh data and you get the Gaussian likelihood — the obvious, principled way to score an estimate. In high dimensions the obvious thing falls apart. The likelihood lives or dies on the smallest eigenvalues of the estimate, which are exactly the ones you cannot pin down when you have about as many variables as observations; so, as I will show, it ranks estimators worse than a coin flip. This paper is about a one-knob repair. The Gaussian likelihood factors, through Schur complements, into a sum of block-conditional pieces, and a single dial γ damps how much each block is allowed to trust the others. At $\gamma = 1$ you have the full likelihood; at $\gamma = 0$ the block-diagonal (composite) likelihood that ignores the coupling altogether; in between, something better conditioned than either. I call it the *Schur pseudo-likelihood*. It is the same Schur-complement dial that interpolates hierarchical risk parity and the minimum-variance portfolio — borrowed from allocation and pointed at evaluation instead. I work out what γ does exactly in a tractable case, where the best setting turns out to be the *reliability* of the coupling (a Wiener / James–Stein shrinkage); I show where it helps and where it does not; and I try to be honest about which claims I can actually stand behind.

1 The likelihood quietly fails

If you want to know whether a covariance estimate is any good, the textbook answer is to score it by the likelihood it assigns to held-out data. For a zero-mean Gaussian that is

$$\ell(\widehat{\Sigma}; x) \;=\; -\tfrac{1}{2}\big(\log\det\widehat{\Sigma} + x^\top\widehat{\Sigma}^{-1}x\big),$$

or, averaging over a test sample and writing S for the sample covariance, $\ell(\widehat{\Sigma}; S) = -\frac{1}{2}\big(\log\det\widehat{\Sigma} + \operatorname{tr}(\widehat{\Sigma}^{-1}S)\big)$ — the same thing the Wishart likelihood and Stein's loss are measuring, dressed differently. It is a strictly proper scoring rule. It is what the maximum-likelihood estimator is built to optimize. By every classical argument it is the right answer, and in low dimensions it is.

So here is the awkward part. Both terms above are run by the *smallest* eigenvalues of $\widehat{\Sigma}$: the log-determinant is $\sum_i \log\lambda_i$, which dives toward $-\infty$ as any $\lambda_i \to 0$, and $\widehat{\Sigma}^{-1}$ blows up in the same limit. When the dimension p is comparable to the sample size n — the regime everyone actually works in — those small eigenvalues collapse toward zero and, worse, become statistically unidentifiable: the Marčenko–Pastur bulk is noise pretending to be signal. The likelihood then stakes its entire verdict on the part of the spectrum it knows least about. You can guess how that goes.

It is not subtle. In a controlled experiment where I know which of two estimates is genuinely better (I can compute the true Kullback–Leibler divergence, because I built the truth), the held-out likelihood picks the better one about 45% of the time. A coin would do better. Inversion-free judges — Frobenius distance, the variogram score — sail along at around 85%. The likelihood, the one estimator we were told to trust, is the one that fails. This is the evaluation-side twin of Markowitz's "error maximization": the minimum-variance portfolio $w \propto \widehat{\Sigma}^{-1}\mathbf{1}$ falls apart for the very same reason, the very same $\widehat{\Sigma}^{-1}$, the very same small eigenvalues. The cure, in both places, is to stop trusting the raw inverse. The question is how to stop trusting it *by degrees*, with one interpretable knob, rather than throwing the coupling away entirely.

2 What the Schur complement is doing here

Nothing in this paper works without one exact fact, so let me state it carefully before I start bending it. Split the variables into K ordered blocks, $x = (x_1, \ldots, x_K)$. The joint density factors — exactly, no approximation — into a product of block conditionals, $p(x) = \prod_k p(x_k \mid x_{<k})$, and for a Gaussian each conditional is itself Gaussian with covariance the *Schur complement*

$$S_k \;=\; \Sigma_{kk} - \Sigma_{k,<k}\, \Sigma_{<k,<k}^{-1}\, \Sigma_{<k,k}$$

and mean $\mu_{k|<k} = \Sigma_{k,<k}\Sigma_{<k,<k}^{-1}x_{<k}$. So $\log\det\Sigma = \sum_k \log\det S_k$, the quadratic form splits block by block, and the whole likelihood is just a sum of block-conditional terms, each of which only ever inverts the conditioning block $\Sigma_{<k,<k}$. This is the block Cholesky factorization, and it is the engine underneath the Vecchia approximation, Gaussian–Markov-random-field likelihoods, and nested-dissection elimination. I have made no approximation; I have only rewritten the likelihood in a form that lets me reach in and turn down one thing.

3 One dial

Here is the dial. Damp the Schur complement, and the regression that comes with it, by a single number $\gamma \in [0, 1]$:

$$S_k(\gamma) \;=\; \Sigma_{kk} - \gamma\, \Sigma_{k,<k}\Sigma_{<k,<k}^{-1}\Sigma_{<k,k} \;=\; (1-\gamma)\,\Sigma_{kk} + \gamma\, S_k, \qquad \mu_{k|<k}(\gamma) = \gamma\, \Sigma_{k,<k}\Sigma_{<k,<k}^{-1}x_{<k},$$

and score with the damped conditionals, $\ell_\gamma(\Sigma; x) = \sum_k \log \mathcal{N}\big(x_k;\, \mu_{k|<k}(\gamma),\, S_k(\gamma)\big)$. At $\gamma = 1$ this is the exact joint likelihood again. At $\gamma = 0$ each block is scored under its own unconditional covariance Σ_{kk}, with no reference to the others — a product of block marginals, which is to say composite likelihood in the sense of Besag, Lindsay, and Varin–Reid–Firth. Everywhere in between is the new object, and the only one that changes the block covariances rather than the data: it inverts nothing larger than a conditioning block, and it is better conditioned than the full likelihood precisely because damping lifts the small eigenvalues that sank it. I call it the *Schur pseudo-likelihood*.

Let me be honest about what it is, because the honesty is load-bearing later. At $\gamma = 1$ it is a normalized joint density. At $\gamma = 0$ it is a composite likelihood: a sum of perfectly good component log-densities that does not itself integrate to one. For γ in between, each term is still a genuine Gaussian log-density — a strictly proper score on its own block — but the damped conditionals are not mutually consistent; they are not the conditionals of any single joint Gaussian. So ℓ_γ is a *pseudo*-likelihood, and that is the bargain: I trade the joint-density property for components that stay numerically sane. As we will see in a moment, you must not try to *estimate* a covariance by maximizing it freely; it is a device for *scoring* a covariance, or for damping one you already have.

4 What the dial actually does

I can say exactly what γ does, at least in the smallest case that still has anything to say. Take two scalar blocks: $x_1 \sim \mathcal{N}(0, a)$ and $x_2 = b\, x_1 + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, s)$, so the true law of x_2 given x_1 has slope b and conditional variance s. Maximize the population ℓ_γ over candidate covariances and something slightly unsettling happens. The maximizer recovers the true predictive law for *every* γ — the effective slope comes out to b and the conditional variance to s — but the covariance it implies is inflated, with off-diagonal $\widehat{\Sigma}_{12} = \Sigma^{\star}_{12}/\gamma$. Push γ down far enough and the implied matrix leaves the positive-definite cone altogether. The damping, in other words, gets undone by rescaling: a free fit simply scales its coupling up by $1/\gamma$ to compensate, and for strong enough coupling it scales right past a valid covariance. That is the precise reason ℓ_γ is no good as a free estimation objective, and is fine as a score or as a fixed transform — both of which hold $\widehat{\Sigma}$ still instead of letting it absorb the dial.

Now the part worth the trip. The thing γ is really controlling is how much to trust the estimated coupling, and *that* has a clean optimum. Fit the slope from n samples by least squares; $\hat{b}$ is unbiased with variance $s/(a(n-2))$. Choosing γ to minimize the expected predictive residual gives

$$\boxed{\gamma^\star = \frac{b^2}{b^2 + \operatorname{Var}\hat{b}} = \frac{(n-2)\,\rho^2}{(n-2)\,\rho^2 + (1-\rho^2)}\,, \qquad \rho^2 = \frac{b^2 a}{b^2 a + s}.}$$

Read it and the whole thesis is sitting there. $\gamma^\star$ is the *reliability* of the coupling — a Wiener filter, a James–Stein shrinkage, whatever you like to call the signal-to-total ratio. It climbs to 1 as the sample grows or the coupling strengthens (trust it; use the full likelihood) and falls to 0 as the coupling dissolves into noise (ignore it; go block-diagonal), and it sits in the interior exactly when the coupling is worth something but not everything. I checked it against the simulated optimum and it lands; the vector version replaces ρ^2 with the largest squared canonical correlation between the blocks, and across K blocks the inflation, and the positive-definite threshold, compound down the chain.

5 Why an interior γ wins, and where it doesn't

It pays to be careful here, because the easy story is wrong. Write the score as a quadratic form, $\ell_\gamma(\widehat{\Sigma}; x) = -\frac{1}{2}\big(\mathrm{Ld}_\gamma + x^\top W_\gamma(\widehat{\Sigma})\,x\big)$, where W_γ assembles the damped block conditionals and, at $\gamma = 1$, is exactly $\widehat{\Sigma}^{-1}$. For two fixed estimates and a Gaussian test sample the probability of ranking the better one above the worse is governed by a per-sample signal-to-noise ratio, and that ratio — I worked it out and checked it against Monte Carlo — is *maximized at* $\gamma = 1$. Under test-sample noise alone, Neyman–Pearson wins and damping only adds bias. So the high-dimensional advantage is not a test-noise effect, and any paper that claims otherwise is hand-waving.

Where it comes from is the instability of the *estimate*. The score depends on $\widehat{\Sigma}$ through W_γ, and at $\gamma = 1$, $W_1 = \widehat{\Sigma}^{-1}$ has a sensitivity that scales like $1/\lambda_{\min}^2$; the variance of the score gap across training draws then runs like $1/\lambda_{\min}^4$, which I can watch explode — from essentially nothing to dozens — as the truth gets more ill-conditioned, while a damped γ keeps it bounded. When the discriminating signal lives in the within-block structure and the cross-block coupling is shared noise, an interior γ beats both ends outright (power around 0.84 against 0.45 at either extreme). The dial is a bias–variance knob, but over the *estimate*, not the test point.

One more honest line in the ledger. The estimating equation $\nabla_\Sigma \ell_\gamma$ is unbiased at the truth only at the two ends — $\gamma = 1$ (the Fisher score) and $\gamma = 0$ (correct block marginals); in the interior it is biased, which is the inflation from two sections ago wearing a different hat. So the classical Godambe efficiency story applies only at the endpoints, and the interior is a *tempering*, justified by the discrimination it buys rather than by being any kind of consistent estimator. That is the rigorous form of "score with it, don't fit with it."

6 It is a shrinkage, and a familiar one

The well-behaved use — damp the coupling of a covariance you already have — is just shrinkage, worth saying plainly. Each conditional covariance is pulled from its full Schur complement back toward the unconditional block, $S_k(\gamma) = (1-\gamma)\Sigma_{kk} + \gamma S_k$, with intensity $1-\gamma$. Since conditioning only ever shrinks variance, $S_k \preceq \Sigma_{kk}$, this raises the conditional covariance and lifts the small eigenvalues of every block-sized inverse the score has to perform. It is James–Stein applied blockwise to the conditioning. And it is orthogonal to the shrinkage everyone already knows: Ledoit–Wolf and the rotation-invariant estimators clean the *eigenvalues*, basis-free; this cleans the *cross-block coupling*, basis-dependent. They compose. γ is to the block factorization what shrinkage intensity is to the spectrum.

7 One knob, several familiar faces

The reason I think this is worth a paper rather than a footnote is that the same dial keeps turning up, and naming it organizes a surprising amount of scattered machinery onto two axes of one factorization. The first axis is *which* earlier blocks each block conditions on — the sparsity that Vecchia, GMRFs, banding and nested dissection all regularize along. The second is *how strongly* that conditioning is trusted, which is γ, and which those methods do not have: they run at full trust and economize on structure. The two are orthogonal and compose.

	$\gamma = 1$ (full trust)	$\gamma = 0$ (no trust)
dense conditioning	exact Gaussian likelihood; MVO $w \propto \Sigma^{-1}\mathbf{1}$	block composite likelihood; HRP
sparse conditioning	block-Vecchia / GMRF likelihood	(block-marginal, same column)
interior γ: the Schur pseudo-likelihood — the bridge none of the above carries		

The portfolio row is not an analogy. Minimum-variance optimization is $\gamma = 1$ with full conditioning; hierarchical risk parity is the $\gamma = 0$ corner; Schur complementary allocation interpolates them by damping the very same Schur complement with the very same γ. The Schur pseudo-likelihood is that construction transported from risk to evaluation — identical algebra, new job — and the reason an interior γ beats the minimum-variance endpoint is the reason HRP is steadier than raw mean–variance.

8 It travels

Nothing above needed finance, or even covariance estimation in particular. The ingredients are a Gaussian (or Gaussian-pseudo) block likelihood and a partition, so the coupling-strength dial is generic to Gaussian-block models. In spatial statistics and Gaussian processes, where Vecchia and GMRFs already condition each site on a neighborhood, γ adds a robustness knob on how far to trust those neighbors, orthogonal to the neighborhood choice. In Gaussian graphical models it tempers conditional-dependence strength between the full penalized likelihood and node-wise pseudo-likelihood. In longitudinal and state-space models, with time slices as blocks, it interpolates between treating slices independently and the full temporal joint. And anywhere composite likelihood is already used for tractability, γ is the dial from composite back toward full. I want to be clear that this last paragraph is a claim about the mechanism, not a row of experiments; I have tested the covariance-evaluation case and am pointing at the rest.

9 Geometry, and what actually happened

Damping interpolates each conditional covariance between Σ_{kk} and S_k, and the obvious straight line is not the only road. Moving instead along the affine-invariant SPD geodesic, $\Sigma_{\mathrm{BD}} \#_\gamma \Sigma = \Sigma_{\mathrm{BD}}^{1/2}(\Sigma_{\mathrm{BD}}^{-1/2}\Sigma\,\Sigma_{\mathrm{BD}}^{-1/2})^\gamma \Sigma_{\mathrm{BD}}^{1/2}$, lifts the small eigenvalues multiplicatively rather than additively. The two agree at the ends and to first order, and part ways near a singular Schur complement — which is exactly the strong-coupling regime where it matters: the linear damping shows a power dip there and the geodesic one does not.

As for results, the honest summary is three lines. The optimum in γ moves with the regime: it climbs to 1 when the coupling is trustworthy, falls to 0 when it is noise, and in between an interior γ strictly beats both ends, which is the bias–variance trade made visible. The geodesic variant is the safer default where the two interpolations diverge. And on real equity returns — Ken French portfolios, rolling minimum-variance out-of-sample — the estimators that invert a raw covariance blow up by orders of magnitude while well-conditioned shrinkage stays put; that is not a test of ℓ_γ as an equity estimator, and I will not pretend it is, but it is the same moral the likelihood teaches, paid out in basis points: in high dimensions you are rewarded for not trusting the raw inverse.

10 What I can't yet claim

The closed-form $\gamma^\star$ is real, but it is the *estimator's* optimum — the predictive, well-behaved use. The high-dimensional *discrimination* optimum is only half pinned down: the variance term is first-order and I have it ($1/\lambda_{\min}^4$ at $\gamma = 1$, bounded below it), but the signal term is second-order in the sampling noise — it is the shrinkage benefit — so a closed form for the discrimination $\gamma^\star$ wants the next order, and for now it is characterized only numerically. The compounded K-block positive-definite threshold I have as a two-block formula and as numerical evidence of compounding, not yet as a chain. Block ordering matters, as it does for Vecchia, and I have not chased the permutation question. And the thesis is falsifiable in the right way: the interior optimum should vanish whenever the coupling is fully reliable or pure noise, and an interior optimum in a regime known to be fully reliable would sink it.

Acknowledgements

The original Schur complementary theory, of which this is the evaluation-side mirror, was developed with the support of Intech Investments.

References

López de Prado (2015, 2016); Antonov, Lipton & López de Prado (2024); Cotton (2024, arXiv:2411.05807); Besag (1975); Lindsay (1988); Varin, Reid & Firth (2011); Vecchia (1988); Katzfuss & Guinness (2021); Pan et al. (2024); Rue & Held (2005); George (1973); Ledoit & Wolf (2004, 2012); Bun, Bouchaud & Potters (2017); James & Stein (1961); Gneiting & Raftery (2007); Scheuerer & Hamill (2015). Full annotated lists accompany the `precise` repository.